

UNIT IV

Exact Differential Equation

❖ Show that the following differential equation is exact and hence find it.

1. $(x^2 - 2xy + 3y^2)dx + (y^2 + 6xy - x^2)dy = 0$.
2. $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$.
3. $\frac{dy}{dx} + \frac{y \cos x + \sin y}{\sin x + x \cos y} = 0$.
4. $(x^2 - x + y^2)dx + (-ye^y + 2xy)dy = 0$.
5. $(1 + e^{x/y})dx + (1 - \frac{x}{y})e^{x/y}dy = 0$.
6. $(x^2 - 2xy)dx + (\sin y - x^2)dy = 0$.
7. $(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0$ (Apr/May 2018).
8. $(e^y + y \cos xy)dx + (xe^y + x \cos xy)dy = 0$
9. $(e^y + 1) \cos x dx + e^y \sin x dy = 0$
10. $(xe^{xy} + 2y)\frac{dy}{dx} + ye^{xy} = 0$.
11. $(y \cos x + 1)dx + \sin x dy = 0$ (Model May 2016)

Equation Reducible to exact form

1. $x dx = y(x^2 + y^2 - 1)dy$
2. $y dx - x dy - 3x^2y^2e^{x^3}dx = 0$
3. $(2xy^2 - y)dx + xdy = 0$.
4. $x dx + y dy = a(x^2 + y^2)dy$
5. $x \frac{dy}{dx} - y = x\sqrt{x^2 + y^2}$
6. $y dx - (x - 2y^3)dy = 0$
7. $(x^2 + y^2 + 1)dx - 2xy dy = 0$
8. $x dy - y dx = \sqrt{x^2 + y^2}dx$
9. $(1 + xy)y dx + (1 - xy)x dy = 0$

Rules for finding integrating factor

1. $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$
2. $(x^3y^3 - x^2y^2 + xy + 1)y dx + (x^3y^3 - x^2y^2 - xy + 1)x dy = 0$

First Order Linear Equation

1. $\frac{dy}{dx} + \frac{y}{x} = xe^x$
2. $\frac{1}{x} \frac{dy}{dx} + \frac{y}{x} \tan x = \cos x$
3. $\frac{dy}{dx} + y \cot x = \cos x$
4. $\cos^2 x \frac{dy}{dx} + y = \tan x$
5. $x \frac{dy}{dx} + y = e^x - xy$
6. $(1 + y^2)dx = (\tan^{-1} y - x)dy$
7. $\frac{dy}{dx} + \frac{y}{x} = x^3 - 3$
8. $(x + 1) \frac{dy}{dx} - y = e^x(1 + x)^2$
9. $\frac{dy}{dx} - y \tan x = 3e^{-\sin x}$ if $y = 4$ when $x = 0$

Bernoulli's Equation

1. $x \frac{dy}{dx} + y = x^3 y^6$
2. $\frac{dy}{dx} - y \tan x = \frac{\sin x \cos^2 x}{y^2}$
3. $\tan y \frac{dy}{dx} + \tan x = \cos y \cos^2 x$
4. $\frac{dy}{dx} - 2y \tan x = y^2 \cos^3 x$
5. $xy - \frac{dy}{dx} = y^3 e^{-x^2}$
6. $\frac{dy}{dx} + \frac{x}{1-x^2} y = 3xy^{\frac{1}{2}}$
7. $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 x$
8. $\cos^2 x \frac{dy}{dx} + y = \tan x$
9. $\tan y \frac{dy}{dx} + \tan x = \cos y \cos^2 x$
10. $(x + 1) \frac{dy}{dx} + 1 = 2e^{-y}$ (Model Dec. 2017)

Orthogonal Trajectories (OT)

❖ **Find OT from the following family of curves**

1. Rectangular hyperbola: $xy = c^2$

2. circles: (a) $x^2 + y^2 = a^2$ (b) $r = a \cos \theta$ (c) $r = a(1 - \cos \theta)$
3. Parabola: $y^2 = 4ax$
4. Hypo cycloids: $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$
5. Ellipse: (a) $\frac{x^2}{a^2} + \frac{y^2}{4a^2} = 1$ (b) $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1, \lambda$ is constant
6. Find O.T. of the curves $r^n = a^n \cos n\theta$ and deduce O.T of the series of lemniscates $r^2 = a^2 \cos 2\theta$.
7. Prove that the system of co-focal and co-axial parabola $y^2 = 4a(x + a)$ is self orthogonal.

Differential Equation of 1st order and higher degree

❖ Equation Solvable for p:

1. $p^2 - 7p + 12 = 0$.
2. $\left(\frac{dy}{dx}\right)^2 + 6\left(\frac{dy}{dx}\right) + 8 = 0$.
3. $p^2 - 9p + 18 = 0$.
4. $p^2 + 2py \cot x = y^2$. (Model May 2017, May 2016, Jan 2016)
5. $p^3 + 2xp^2 - y^2p^2 - 2xy^2p = 0$
6. $p^2 - 9p + 2 = 0$
7. $p^3 - 7p - 6 = 0$
8. $p^2 - 2py = 3y^2$
9. $y\left(\frac{dy}{dx}\right)^2 + (x - y)\frac{dy}{dx} - x = 0$.
10. $2p^2 - (x + 2y^2)p + xy^2 = c$ (Model Dec. 2017)

❖ Equation Solvable for x

1. $x = y + a \log p$.
2. $x = y + p^2$.
3. $x = 1 - \frac{p}{\sqrt{p^2 + 1}}$.
4. $p^3 - 4xyp + 8y^2 = 0$.
5. $y = 2px + y^2p^3$ (Model Jan. 2016)
6. $x^2 = 1 + p^2$.
7. $y^2 \log y = xyp + p^2$.

$$8. x = \tan^{-1}(p) + \frac{p}{1+p^2}.$$

$$9. p^2y + 2px - y = 0.$$

$$10. 3p^2e^y - px - 1 = 0.$$

❖ **Equation Solvable for y**

$$1. x - yp = 9p^2.$$

$$2. xp^2 - 2px + x = 0. \text{ (May 2015)}$$

$$3. y = -px + x^4p^2.$$

$$4. y = 3x + \log p.$$

$$5. y = x + a \tan^{-1} p.$$

$$6. 4y = x^2 + p^2.$$

$$7. y = 2px + p^4x^2.$$

$$8. y = -px + x^4p^2.$$

$$9. x + xp^2 + 2yp = 0.$$

$$10. xp^2 + 2px - y = 0.$$

❖ **Clairaut's Equation**

$$1. y = 2px + y^2p^3 \text{ (Model May 2017, May 2015).}$$

$$2. x^2(y - px) = yp^2.$$

$$3. (px - y)(py + x) = 2p.$$

$$4. y = (x - a)p - p^2. \text{ (April/May 2018)}$$

$$5. y = xp + p^2.$$

$$6. (y - px)(p - 1) = p. \text{ (May 2015)}$$

$$7. y = px + ap^{-1}. \text{ (Apr./May 2018)}$$

$$8. y = xp + \sqrt{p^2 + 1}.$$

$$9. y = xp + \frac{ap}{\sqrt{1+p^2}}.$$

$$10. y = x \frac{dy}{dx} + a \frac{dx}{dy}.$$